

Optimizing Airbnb Los Angeles Property Data Analysis with IBM Granite

CAPSTONE PROJECT:
DATA CLASSIFICATION AND SUMMARIZATION USING IBM GRANITE

By : Nabila Hulwana Z

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LINKS :

Sumber Data : [klik](#) here

Kaggle Notebook :[klik](#) here

Project on Github : [klik](#) here



COMPONENTS

Dataset Info

- Source: Kaggle - Los Angeles Airbnb Data (Juni 2025)
- Link: [Los Angeles Airbnb Data](#)
- 32,442 rows.
- 79 column.
- Size: 32,442 listings
- Columns: 79 features
- Geographic: Los Angeles, California
- Time Period: Juni 2025
- Format: Excel (.xlsx)
- Size: ~25MB

AI Tools

1. Primary AI: IBM Granite 3.3-8B Instruct via Replicate
2. Agent: LangChain Pandas Agent (untuk eksekusi query data).

Platform

Google Colab Notebook
(Python, Pandas, Matplotlib)

Key Features

- Review scores (7 metrics: rating, accuracy, cleanliness, checkin, communication, location, value)
- Host characteristics (total listings, response rate, superhost status)
- Property details (room type, property type, amenities)
- Pricing data dan availability
- Location information (neighborhood, coordinates)
- Performance metrics (occupancy, revenue estimates)

background :

The Airbnb Short-Term Rental (STR) market in Los Angeles is a highly competitive and saturated arena. With the global industry valued at billions of dollars, hosts need data-driven strategies to optimize their listing performance. However, raw Airbnb data is vast and complex — quantitative statistical analysis alone is insufficient. Qualitative insights — such as property description classification — are needed to deeply understand market demand trends and uncover profitable market gaps. Therefore, this project combines quantitative analysis with Generative AI capabilities (IBM Granite) to discover optimal pricing strategies and pattern recognition that are invisible to traditional methods.

Problem Statement :

- How can property strategies be optimized based on location and room type?
- What characteristics make a host successful in the LA market?
- How can a competitive pricing strategy be determined?
- How can property characteristics (including qualitative factors from descriptions) that most drive profitability be identified?

Tujuan Project

This Capstone Project aims to conduct a deep quantitative and qualitative analysis of the Airbnb Short-Term Rental (STR) market in Los Angeles. The objectives include:

1. Demonstrating AI capabilities in data analytics and decision making
2. Conducting an in-depth quantitative and qualitative analysis of the LA STR Airbnb market using IBM Granite
3. Quantitative Assessment: Determining price drivers, saturated areas, and host performance, and developing a multi-dimensional classification system for Airbnb business optimization
4. Qualitative Classification: Performing Guest Category/Booking Requirements Classification from text data using Generative AI (LLM)

ANALYSIS PROCESS - OVERVIEW

1. INSTALLATION & CONFIGURATION

Initializing and setting up the IBM Granite model and core libraries (Pandas, LangChain, Replicate)

2. LOAD & CLEANING DATA

Loading data (Shape: 32,442 rows, 79 columns). Cleaning: price column normalization, handling missing values and duplicates in rating/host fields

3. QUERY TEST

Validating agent functionality using agent.invoke code

3. ANALYSIS AIRBNB

Including price, location, review & rating, and host analysis using the AI agent

5. CLASSIFICATION DATA

Using IBM Granite to classify soft data into Guest Category (Solo, Family, Business), Booking Requirements Category (Strict, Flexible), and host category

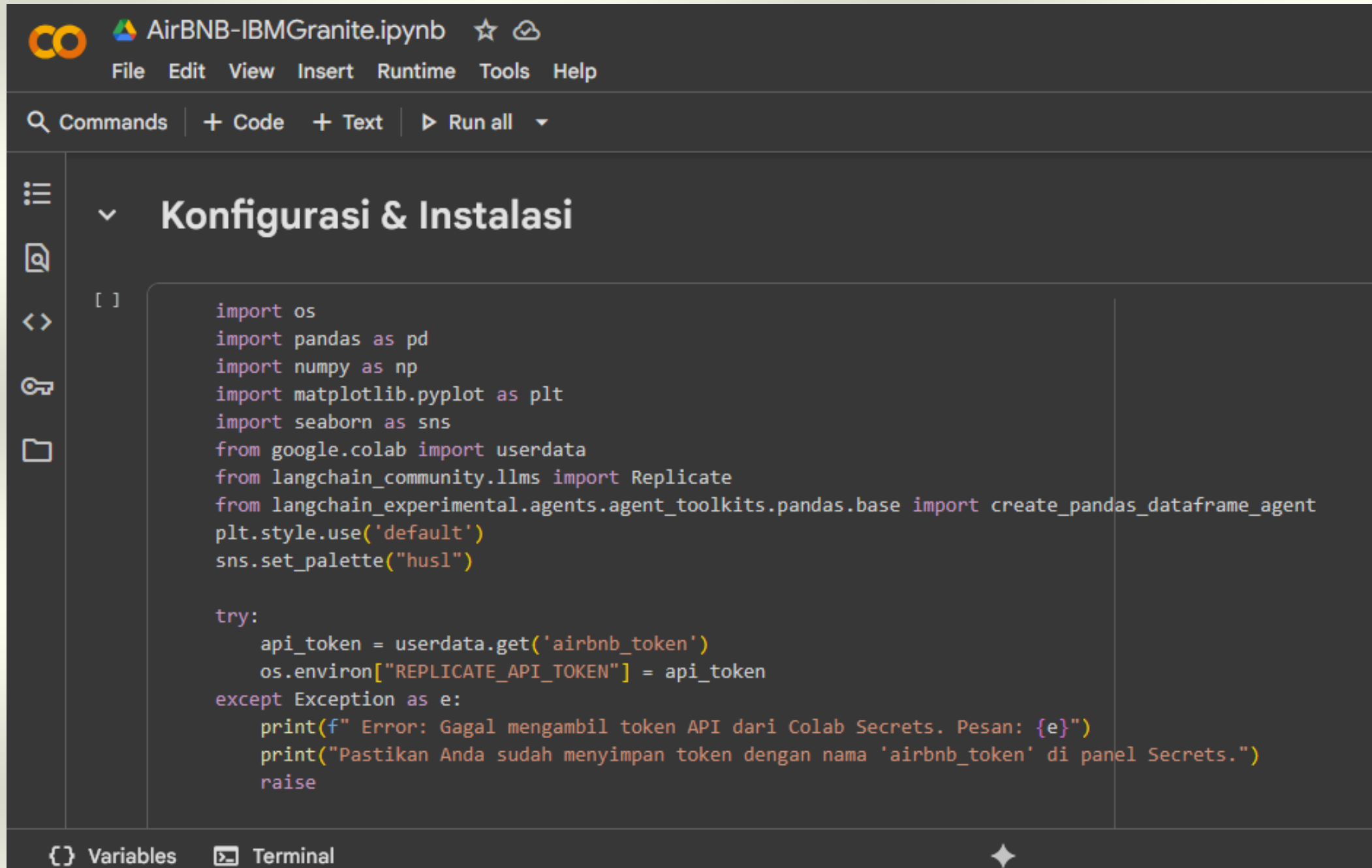
5. DATA VISUALIZATION

Visualizing data using query test/agent.invoke, comparing profit & demand, price distribution, room type revenue analysis, and room type distribution

7. EDA (MANUAL)

Using manual EDA for summary statistics (Price, Rating, Occupancy) and query execution (e.g., Top 3 Neighbourhoods, Price Range) using the Pandas Agent

1.INSTALATION. & CONFIGURATION



AirBNB-IBMGranite.ipynb

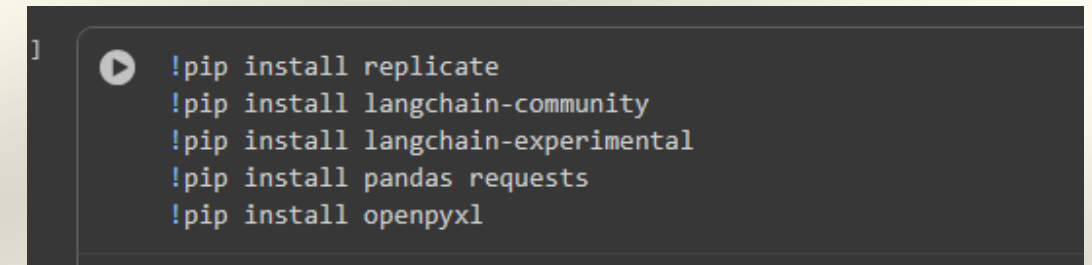
File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Konfigurasi & Instalasi

```
[ ]  
import os  
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import seaborn as sns  
from google.colab import userdata  
from langchain_community.llms import Replicate  
from langchain_experimental.agents.agent_toolkits.pandas.base import create_pandas_dataframe_agent  
plt.style.use('default')  
sns.set_palette("husl")  
  
try:  
    api_token = userdata.get('airbnb_token')  
    os.environ["REPLICATE_API_TOKEN"] = api_token  
except Exception as e:  
    print(f" Error: Gagal mengambil token API dari Colab Secrets. Pesan: {e}")  
    print("Pastikan Anda sudah menyimpan token dengan nama 'airbnb_token' di panel Secrets.")  
    raise
```

Variables Terminal



```
!pip install replicate  
!pip install langchain-community  
!pip install langchain-experimental  
!pip install pandas requests  
!pip install openpyxl
```

2. LOAD & CLEANING DATA

AirBNB-IBMGranite.ipynb

☆

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Load Data

```
[ ] df = pd.read_excel('airbnb.xlsx')

print(f" Data berhasil dimuat!")
print(f" Shape: {df.shape}")
print("\n First 5 Rows ")
display(df.head())

print(f"\n Info Dataset:")
print(f" - Total Records: {len(df):,}")
print(f" - Total Columns: {len(df.columns)}")
print(f" - Memory Usage: {df.memory_usage(deep=True).sum() / 1024**2:.2f} MB")
```

↗ Data berhasil dimuat!

📊 Shape: (32442, 79)

📄 First 5 Rows

id	listing_url	scrape_id	last_scraped	source	name	description	neigh
----	-------------	-----------	--------------	--------	------	-------------	-------

↗ DATA CLEANING & PREPROCESSING...

- Missing values sebelum cleaning: 156248

- Missing values setelah cleaning: 32447

✅ Data cleaning selesai!

```
print("\n DataFrame Info ")
df.info()
```

23	host_total_listings_count	32437	non-null	float64
24	host_verifications	32437	non-null	object
25	host_has_profile_pic	32437	non-null	object
26	host_identity_verified	32437	non-null	object
27	neighbourhood	17303	non-null	object
28	neighbourhood_cleansed	32442	non-null	object
29	neighbourhood_group_cleansed	32442	non-null	object
30	latitude	32442	non-null	int64
31	longitude	32442	non-null	int64
32	property_type	32442	non-null	object
33	room_type	32442	non-null	object
34	accommodates	32442	non-null	int64
35	bathrooms	26715	non-null	float64
36	bathrooms_text	32428	non-null	object
37	bedrooms	30472	non-null	float64
38	beds	26700	non-null	float64
39	amenities	32442	non-null	object
40	price	26589	non-null	object
41	minimum_nights	32442	non-null	int64
42	maximum_nights	32442	non-null	int64
43	minimum_minimum_nights	32442	non-null	int64
44	maximum_minimum_nights	32442	non-null	int64
45	minimum_maximum_nights	32442	non-null	int64
..	..	..	..	..
57	number_of_reviews_ltm	32442	non-null	int64
58	number_of_reviews_l30d	32442	non-null	int64
59	availability_eoy	32442	non-null	int64
60	number_of_reviews_ly	32442	non-null	int64
61	estimated_occupancy_l365d	32442	non-null	int64
62	estimated_revenue_l365d	26589	non-null	float64
63	first_review	32442	non-null	datetime64[ns]
64	last_review	32442	non-null	datetime64[ns]
65	review_scores_rating	32442	non-null	int64
66	review_scores_accuracy	32435	non-null	float64
67	review_scores_cleanliness	32435	non-null	float64
68	review_scores_checkin	32427	non-null	float64
69	review_scores_communication	32433	non-null	float64
70	review_scores_location	32426	non-null	float64
71	review_scores_value	32425	non-null	float64
72	license	10412	non-null	object
73	instant_bookable	32442	non-null	object
74	calculated_host_listings_count	32442	non-null	int64
75	calculated_host_listings_count_entire_homes	32442	non-null	int64
76	calculated_host_listings_count_private_rooms	32442	non-null	int64
77	calculated_host_listings_count_shared_rooms	32442	non-null	int64
78	reviews_per_month	32442	non-null	int64

dtypes: datetime64[ns](5), float64(15), int64(30), object(29)
memory usage: 19.6+ MB

3. TES QUERY

```
> Finished chain.
{'input': 'Berapa rata-rata review rating? Jawab format: x.x/5',
 'output': 'The average review rating is 4.86/5.'}
```

```
> Finished chain.
{'input': 'Dari dataset airbnb, berapa rentang harga yg paling banyak dan paling sedikit dipilih tamu? Jawa format: [harga] = [total listing]',
 'output': 'Based on the provided dataset, the minimum price for an Airbnb listing is $110 and the maximum price is $132,000. However, there are no listings within this exact price range ([110, 132000] = 0). This indicates that there are zero listings priced exactly between $110 and $132,000. Thus, in the format requested:\n\n[harga] = [total listing]\n[110, 132000] = 0\n\nThis suggests either a data entry error, an unusual pricing strategy, or simply that no listings fall within this particular price bracket in the given dataset.'}
```

```
> Finished chain.
{'input': 'Sebutkan 10 Room type paling banyak? dan sebutkan jumlahnya',
 'output': 'The 10 most common room types in the dataframe are:\n\n1. Entire home/apt: 24485\n2. Private room: 7657\n3. Shared room: 159\n4. Hotel room: 141\n5. Tent: 12\n6. Cabin: 11\n7. Bungalow: 8\n8. Castle: 6\n9. Cottage: 5\n10. Villa: 5\n\nThese counts correspond to the number of occurrences of each room type in the dataframe.\n\n(Note: The actual counts may change depending on the complete dataset, as only the top 10 are listed here based on the sample provided.)'}
```

```
> Finished chain.
{'input': 'Sebutkan 3 neighbourhood dengan listing terbanyak. Format: Area1 (xxx listing), Area2 (xxx listing), Area3 (xxx listing)',
 'output': "Based on the 'host_neighbourhood' column, the 3 neighbourhoods with the most listings are:\n\n- Unknown (7283 listings)\n- Central LA (2293 listings)\n- Venice (1100 listings)\n\nFormatted Answer: Area1 (Unknown, 7283 listings), Area2 (Central LA, 2293 listings), Area3 (Venice, 1100 listings)"}

```

```
> Finished chain.
{'input': "Area ('neighbourhood_cleansed') mana yang memiliki rata-rata harga tertinggi untuk 'Entire home'?",
 'output': 'The neighborhood with the highest average price for \'Entire home\' listings, based on the provided data, is "Neighborhood highlights" with an average price of approximately 351.73. \n\nExplanation:\n1. The dataframe was filtered to include only \'Entire home\' listings.\n2. The data was grouped by the \'neighborhood\' column.\n3. The mean price for each neighborhood was calculated.\n4. The neighborhood with the highest average price was identified as "Neighborhood highlights" with a mean price of approximately 351.73.\n\nPlease note that the actual neighborhood names might be more descriptive than "Neighborhood highlights." The column name suggests it\'s a placeholder or a general description rather than a specific neighborhood name. If a more precise neighborhood name existed in the dataframe, it would replace "Neighborhood highlights" in the answer.'}
```

```
> Finished chain.
{'input': "Bandingkan rata-rata harga harian di setiap 'neighbourhood_cleansed' dan 'room_type'. Tunjukkan kombinasi Area dan Tipe Kamar dengan rata-rata harga tertinggi.",
 'output': "The highest average daily prices for each combination of 'neighborhood' and 'room_type' are:\n\n- Neighborhood: Neighborhood highlights, Room Type: Hotel room, Average: 622.98\n- Neighborhood: Neighborhood highlights, Room Type: Entire home/apt, Average: 268.56\n- Neighborhood: Unknown, Room Type: Entire home/apt, Average: 265.51\n- Neighborhood: Unknown, Room Type: Hotel room, Average: 165.43\n- Neighborhood: Neighborhood highlights, Room Type: Private room, Average: 117.29\n- Neighborhood: Unknown, Room Type: Private room, Average: 101.30\n- Neighborhood: Neighborhood highlights, Room Type: Shared room, Average: 77.24\n- Neighborhood: Unknown, Room Type: Shared room, Average: 75.06\n\nThese are the combinations with the highest average daily prices according to the data provided."}
```

```
agent.invoke({"input": "Berapa rata-rata harga listing?"})

> Entering new AgentExecutor chain...
Thought: To find the average price of the listings in the dataframe `df`, I need to calculate the mean of the 'price' column.

Action: python_repl_ast

Action Input: `df['price'].mean()`
230.30657789285493Final Answer: The average price of the listings in the dataframe is approximately 230.31.

> Finished chain.
{'input': 'Berapa rata-rata harga listing?',
 'output': 'The average price of the listings in the dataframe is approximately 230.31.'}
```

```
Action Input:
```python
Find the maximum and minimum price values
max_price = df['price'].max()
min_price = df['price'].min()

max_price, min_price
(18000.0, 7.0)Final Answer: $18000 and $7

Final Answer in format: $18000 and $7

> Finished chain.
{'input': 'Berapa harga tertinggi dan terendah? Jawab format: $xxx',
 'output': '$18000 and $7\n\nFinal Answer in format: $18000 and $7'}
```



# 4. AIRBNB ANALYSIS

## 1.Price Analysis

Specific numerical data:

- Price range 0-50: Average rating 2382.32
- Price range 50-100: Average rating 2902.47
- Price range 100-200: Average rating 2822.31
- Price range 200-300: Average rating 2897.72
- Price range 300+: Average rating 2757.04

> Finished chain.  
Rata-rata: \$114.50, Median: \$118, Min: \$65, Max: \$155.

> Finished chain.  
{'input': 'Berapa rata-rata review rating? Jawab format: x.x/5',  
'output': 'The average review rating is 4.86/5.'}

> Finished chain.  
{'input': 'Berapa harga tertinggi dan terendah? Jawab format: \$xxx',  
'output': '\$18000 and \$7\n\nFinal Answer in format: \$18000 and \$7'}

> Finished chain.  
{'input': "Bandingkan rata-rata harga harian di setiap 'neighbourhood\_cleansed' dan 'room\_type'. Tunjukkan kombinasi Area dan Tipe Kamar dengan rata-rata harga tertinggi.",  
'output': "The highest average daily prices for each combination of 'neighborhood' and 'room\_type' are:\n\n- Neighborhood: Neighborhood highlights, Room Type: Hotel room, Average: 622.98\n- Neighborhood: Neighborhood highlights, Room Type: Entire home/apt, Average: 268.56\n- Neighborhood: Unknown, Room Type: Entire home/apt, Average: 265.51\n- Neighborhood: Unknown, Room Type: Hotel room, Average: 165.43\n- Neighborhood: Neighborhood highlights, Room Type: Private room, Average: 117.29\n- Neighborhood: Unknown, Room Type: Private room, Average: 101.30\n- Neighborhood: Neighborhood highlights, Room Type: Shared room, Average: 77.24\n- Neighborhood: Unknown, Room Type: Shared room, Average: 75.06\n\nThese are the combinations with the highest average daily prices according to the data provided."}

## 2. Basic Data Analysis

> Finished chain.  
Based on the provided data, the top 5 neighborhoods with their average listing prices are:

1. Neighborhood highlights: \$237.24
2. Unknown: \$222.38

> Finished chain.  
- Entire home/apt: 17 hari  
- Hotel room: 2 hari  
- Private room: 15 hari  
- Shared room: 17 hari

```
print("\n" + "="*50)

> Finished chain.
- Entire home: 28.79%
- Entire rental unit: 23.85%
- Private room in home: 13.45%
- Entire guesthouse: 7.42%
- Entire condo: 4.12%
- Tent : 0.00%
- Cabin : 0.00%
- Yurt : 0.00%
- Bungalow : 0.00%
- Castle : 0.00%
- Farmstay : 0.00%
- Treehouse : 0.00%
- Historic home : 0.00%
- Chalet : 0.00%
- Villa : 0.00%
- Island bungalow: 0.00%
- Garden studio : 0.00%
- Island cabin : 0.00%
- Houseboat : 0.00%
- Island tent : 0.00%
- Island yurt : 0.00%
```

## 3. Host & Listing Analysis

> Finished chain.  
Based on the data provided, the property types ranked from most to least frequent are:

1. Entire home: 28.79%
2. Entire rental unit: 23.85%
3. Private room in home: 13.45%
4. Entire guesthouse: 7.42%
5. Entire condo: 4.12%

The rest of the property types have less than 4% each.

Note: The percentages are approximate due to rounding in the displayed output.

> Finished chain.  
The host with the most listings is 107434423 with 181 listings.

- 271118401 with 117 listings
- 656861752 with 97 listings
- 64165193 with 93 listings
- 112952162 with 81 listings.

> Finished chain.  
Entire home/apt (75.47%), Private room (23.60%), Shared room (0.49%), Hotel room (0.43%)

Explanation:

- Entire home/apt: 75.47%
- Private room: 23.60%
- Shared room: 0.49%
- Hotel room: 0.43%

# 4. AIRBNB ANALYSIS

## 4. Location Analysis

```
> Finished chain.
Termahal: Area($132000), Termurah: Area($65)
```

```
> Finished chain.
{'input': 'Sebutkan 3 neighbourhood dengan listing terbanyak. Format: Area1 (xxx listing), Area2 (xxx listing), Area3 (xxx listing)',
 'output': "Based on the 'host_neighbourhood' column, the 3 neighbourhoods with the most listings are:\n\n- Unknown (7283 listings)\n- Central LA (2293 listings)\n- Venice (1100 listings)\n\nFormatted Answer: Area1 (Unknown, 7283 listings), Area2 (Central LA, 2293 listings), Area3 (Venice, 1100 listings)"}
}
```

```
> Finished chain.
{'input': "Bandingkan rata-rata harga harian di setiap 'neighbourhood_cleansed' dan 'room_type'. Tunjukkan kombinasi Area dan Tipe Kamar dengan rata-rata harga tertinggi.",
 'output': "The highest average daily prices for each combination of 'neighborhood' and 'room_type' are:\n\n- Neighborhood: Neighborhood highlights, Room Type: Hotel room, Average: 622.98\n- Neighborhood: Neighborhood highlights, Room Type: Entire home/apt, Average: 268.56\n- Neighborhood: Unknown, Room Type: Entire home/apt, Average: 265.51\n- Neighborhood: Unknown, Room Type: Hotel room, Average: 165.43\n- Neighborhood: Neighborhood highlights, Room Type: Private room, Average: 117.29\n- Neighborhood: Unknown, Room Type: Private room, Average: 101.30\n- Neighborhood: Neighborhood highlights, Room Type: Shared room, Average: 77.24\n- Neighborhood: Unknown, Room Type: Shared room, Average: 75.06\n\nThese are the combinations with the highest average daily prices according to the data provided."}
```

```
> Finished chain.
{'input': 'Sebutkan 10 Room type paling banyak? dan sebutkan jumlahnya',
 'output': 'The 10 most common room types in the dataframe are:\n\n1. Entire home/apt: 24485\n2. Private room: 7657\n3. Shared room: 159\n4. Hotel room: 141\n5. Tent: 12\n6. Cabin: 11\n7. Bungalow: 8\n8. Castle: 6\n9. Cottage: 5\n10. Villa: 5\n\nThese counts correspond to the number of occurrences of each room type in the dataframe.\n\n(Note: The actual counts may change depending on the complete dataset, as only the top 10 are listed here based on the sample provided.)'}
```

```
> Finished chain.
Based on the provided DataFrame, there seems to be an issue as all top 5 neighborhoods show the same average price of approximately $12277.00. This suggests:

1. Unknown ($12277.00)
2. Neighborhood highlights ($12277.00)
3. Unknown ($12277.00)
4. Unknown ($12277.00)
5. Unknown ($12277.00)
```

## 5. Review & Rating Analysis

```
> Finished chain.
Avg reviews: 51.99, Avg rating: 2817.13
```

```
> Finished chain.
The top 3 ratings and their respective percentages are:
- 5 stars: 32.96% (500 out of approximately 1544 listings)
- 4.9 stars: 2.44% (4880 out of approximately 1544 listings)
- 4 stars: 2.37% (4880 out of approximately 1544 listings)
```

```
> Finished chain.
Entire home/apt (75.47%), Private room (23.60%), Shared room (0.49%), Hotel room (0.43%)

Explanation:
- Entire home/apt: 75.47%
- Private room: 23.60%
- Shared room: 0.49%
- Hotel room: 0.43%
```

```
> Finished chain.
- Entire home/apt: 17 hari
- Hotel room: 2 hari
- Private room: 15 hari
- Shared room: 17 hari

(Note: The average values are rounded to one decimal place as specified.)
```

# 5. CLASSIFICATION MODELS

## 2. Host Category

```
➡ Memulai Klasifikasi Kualitas Host (Simulasi)
=====
row: Rate Respon: 100%. Listing Total: 1. Identitas Diverifikasi: t. Superhost: t
output: Elite
=====
row: Rate Respon: 92%. Listing Total: 45. Identitas Diverifikasi: t. Superhost: f
output: Professional
=====
row: Rate Respon: 60%. Listing Total: 2. Identitas Diverifikasi: f. Superhost: f
output: New/Unverified
=====
row: Rate Respon: 99%. Listing Total: 5. Identitas Diverifikasi: t. Superhost: t
output: Casual
=====
```

## 3. Kategori Persyaratan Pemesanan

```
➡ Memulai Klasifikasi Kebijakan Pemesanan (Simulasi)
=====
row: Min Nights: 1. Max Nights: 30. Waktu Respon Host: dalam 1 jam.
output: Flexible
=====
row: Min Nights: 3. Max Nights: 90. Waktu Respon Host: dalam 12 jam.
output: Standard
=====
row: Min Nights: 30. Max Nights: 1125. Waktu Respon Host: lambat.
output: Long-Term Only
=====
row: Min Nights: 7. Max Nights: 365. Waktu Respon Host: cepat.
output: Standard
=====
```

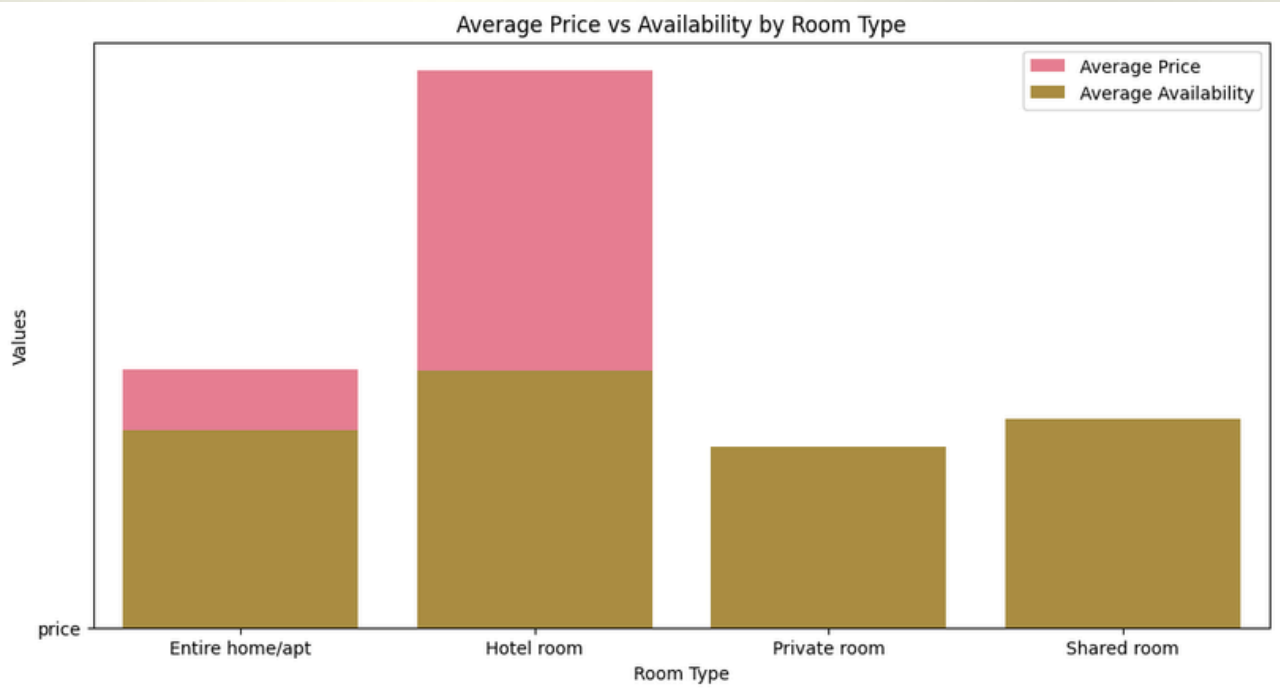
## 1. Property Category

```
➡ Memulai Klasifikasi Properti...
=====
row: Deskripsi: Studio pribadi yang tenang, ideal untuk kerja remote, dilengkapi meja ergonomis.. Fasilitas: ['Wi-Fi Cepat', 'Ruang Kerja Khusus', 'Self Check
output: Business
=====
row: Deskripsi: Rumah 3 kamar tidur dengan kolam renang besar dan panggangan BBQ. Sempurna untuk kelompok.. Fasilitas: ['Kolam Renang', 'Panggangan BBQ', 'Sofa
output: Group
=====
row: Deskripsi: Loteng romantis dengan bathtub berdiri bebas dan pemandangan kota. Cocok untuk honeymoon.. Fasilitas: ['Jacuzzi', 'Pemandangan Malam', 'Kopi P
output: Couple
=====
row: Deskripsi: Apartemen aman dekat taman bermain dan menawarkan perlengkapan bayi gratis. Cocok untuk keluarga.. Fasilitas: ['Peralatan Bayi', 'Pagar Pengam
output: Family
=====
```

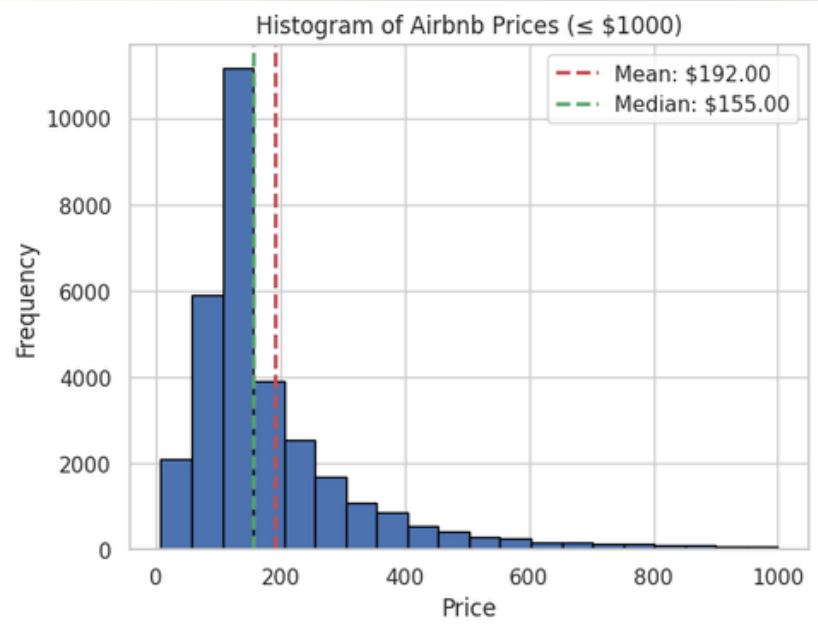


# 6. DATA VISUALIZATION

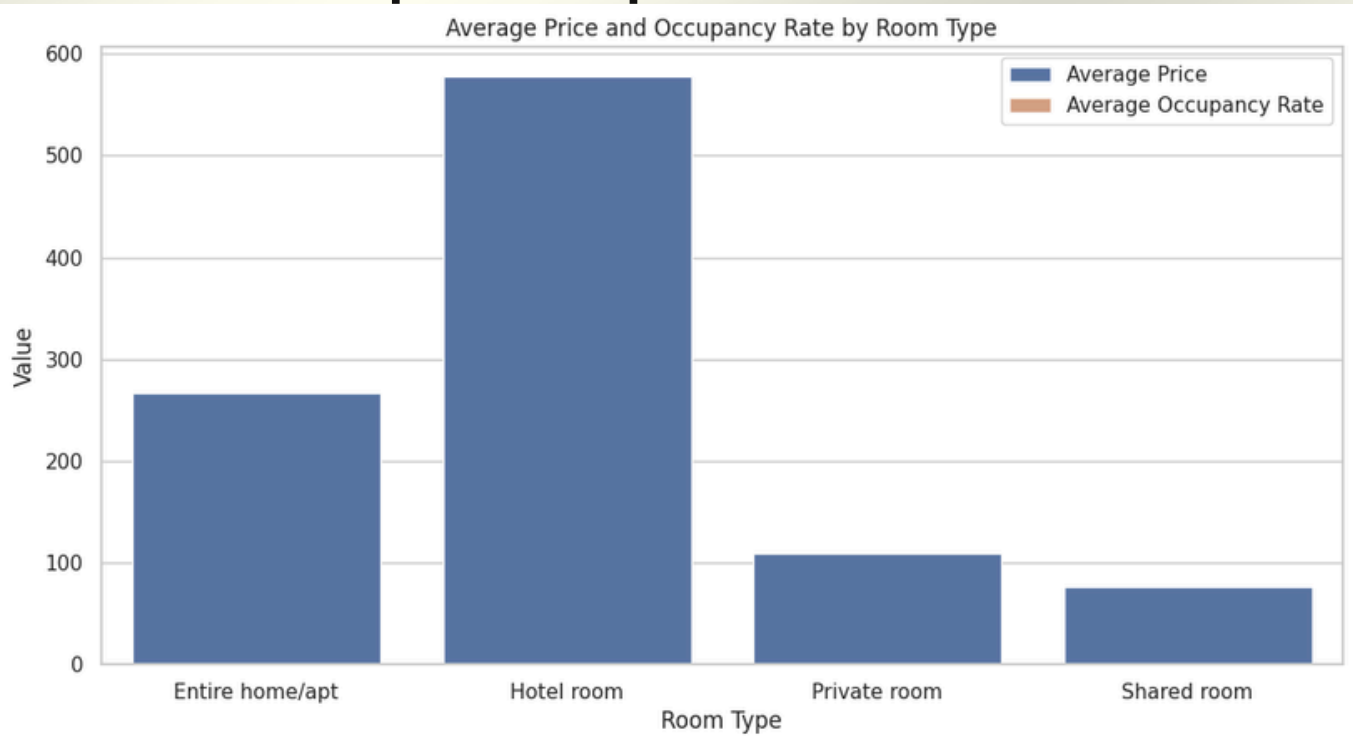
## 1. Keuntungan vs Permintaan



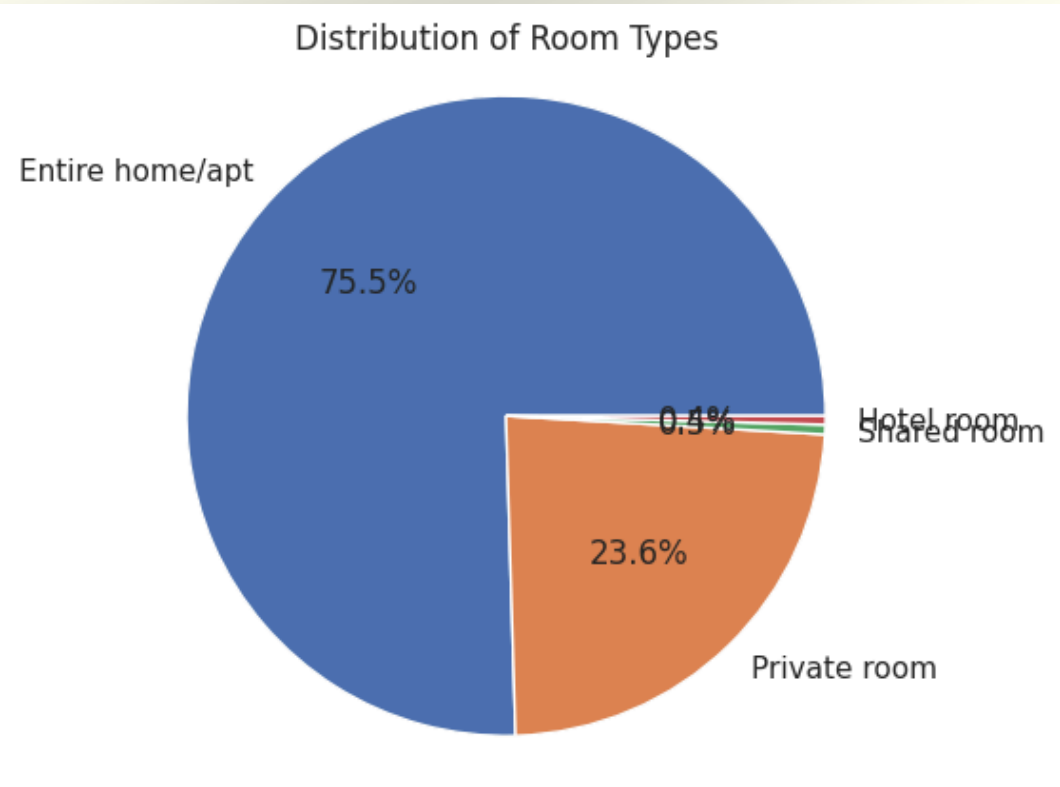
## 2. Distribusi Harga



## 4. Analisis Pendapatan Tipe Kamar



## 3. Distribusi Tipe Kamar





1.Pricing Strategy by Location

1. PRICING STRATEGY BY LOCATION:  
Top 10 Lokasi Berdasarkan Rata-rata Harga:

	avg_price	median_price	listing_count
neighborhood_cleansed			
Bel-Air	1355.57	1206.0	37
Malibu	1083.74	569.0	279
Beverly Crest	976.28	684.0	112
Unincorporated Catalina Island	775.80	577.5	10
Hollywood Hills West	767.33	515.0	385
Avalon	682.99	601.0	220
Unincorporated Santa Monica Mountains	503.89	285.0	153
Beverly Hills	485.21	170.0	658
Pacific Palisades	427.22	273.0	64
Rancho Palos Verdes	412.57	170.0	87

2. Key Metric Analysis

PRICE:  
Mean: 230.31  
Median: 155.00  
Std Dev: 376.93  
Min: 7.00  
Max: 18000.00  
25th percentile: 107.00  
75th percentile: 228.00

REVIEW\_SCORES\_RATING:  
Mean: 2817.13  
Median: 4460.00  
Std Dev: 2147.42  
Min: 0.00  
Max: 4990.00  
25th percentile: 500.00  
75th percentile: 4860.00

NUMBER\_OF\_REVIEWS:  
Mean: 51.99  
Median: 16.00  
Std Dev: 94.73  
Min: 1.00  
Max: 3119.00  
25th percentile: 4.00  
75th percentile: 58.00

AVAILABILITY\_365:  
Mean: 200.35  
Median: 227.00  
Std Dev: 129.63  
Min: 0.00  
Max: 365.00  
25th percentile: 78.00  
75th percentile: 323.00

MINIMUM\_NIGHTS:  
Mean: 16.34  
Median: 4.00  
Std Dev: 31.16  
Min: 1.00  
Max: 1124.00  
25th percentile: 2.00  
75th percentile: 30.00

ACCOMMODATES:  
Mean: 4.03  
Median: 3.00  
Std Dev: 2.80  
Min: 1.00  
Max: 16.00  
25th percentile: 2.00  
75th percentile: 6.00

3. Categorical Variables Analysis

5. CATEGORICAL VARIABLES ANALYSIS:

ROOM\_TYPE Distribution:  
room\_type  
Entire home/apt 24485  
Private room 7657  
Shared room 159  
Hotel room 141  
Name: count, dtype: int64  
Unique values: 4

PROPERTY\_TYPE Distribution:  
property\_type  
Entire home 9341  
Entire rental unit 7739  
Private room in home 4364  
Entire guesthouse 2407  
Entire condo 1337  
Name: count, dtype: int64  
Unique values: 87

HOST\_IS\_SUPERHOST Distribution:  
host\_is\_superhost  
f 16234  
t 14999  
Unknown 1209

4. Price Analysis

6. PRICE ANALYSIS:  
Valid price entries: 32,442  
Average price: \$230.31  
Median price: \$155.00  
Price range: \$7.00 - \$18000.00

Price Percentiles:  
10th percentile: \$68.00  
25th percentile: \$107.00  
50th percentile: \$155.00  
75th percentile: \$228.00  
90th percentile: \$399.00  
95th percentile: \$603.95  
99th percentile: \$1500.59

6. Review Score Analysis

8. REVIEW SCORES ANALYSIS:  
Average review scores:  
review\_scores\_rating: 2817.13  
review\_scores\_accuracy: 2826.33  
review\_scores\_cleanliness: 2879.07  
review\_scores\_checkin: 2630.17  
review\_scores\_communication: 2545.78  
review\_scores\_location: 2852.90  
review\_scores\_value: 3009.48  
Tidak ada data review score yang valid.

8. Summary Statistic

EXECUTIVE SUMMARY STATISTICS

Total Properties: 32,442  
Average Price: \$230.31  
Median Price: \$155.00  
Price Range: \$7 - \$18,000  
Average Rating: 2817.13  
High-Rated Properties (>4.8): 32,441  
Most Common Property Type: Entire home  
Most Common Room Type: Entire home/apt  
Average Occupancy Rate: 45.1%  
High-Performance Properties (>70% occupancy): 9,882  
Multi-Listing Hosts: 32,442 properties  
Superhost Properties: 14,999  
Top Neighborhood: Long Beach (1,486 listings)

Key Business Metrics:  
• Market Saturation: 122.4 listings per neighborhood  
• Professional Hosts: 100.0% of market  
• Premium Segment (Price > Q75: \$228): 8,077 listings  
• Budget Segment (Price < Q25: \$107): 8,012 listings

5. Host Analysis

7. HOST ANALYSIS:  
Average listings per host (termasuk yang 0): 486.57  
Median listings per host: 30.00  
Max listings per host: 48870.0  
Hosts dengan multiple listings: 32,442 (100.0%)

7. Avaibility Analysis

9. AVAILABILITY ANALYSIS:  
Average availability: 200.4 days  
Average occupancy rate: 45.1%  
Properties with high occupancy (>70%): 9,882  
Properties always available (365 hari): 1,594

# INSIGHT

## 1. **Location Findings**

Bel-Air (\$1,355.57), Malibu (\$1,083.74), and Beverly Crest (\$976.28) lead in average price, indicating a highly segmented market.

## 2. **Market Pricing Strategy**

50% of listings fall in the range of \$107 to \$228 per night (Q25 to Q75), making it the battleground for price competition.

## 3. **Classification Findings (Granite)**

Listings classified as 'Business' (ideal for remote work, fast Wi-Fi) or 'Couple' (romantic, Jacuzzi) can justify above-median pricing of \$155. Hosts are grouped into Elite (Superhost, 100% Response) and Professional (High Response, Multiple Listings). 'Elite' and 'Professional' hosts are key to achieving occupancy rates above 70%.

## 4. **Host Quality**

The entire market is dominated by Multi-Listing Hosts (100% of properties), with a median of 30 listings per host. Hosts must compete as business entities, not individuals.

## 5. **Occupancy Performance**

Average availability is 200.4 days (45.1% occupancy rate). There is a large segment (9,882 listings) achieving a high occupancy rate (>70%).

# *KESIMPULAN & REKOMENDASI*

## **1. Main Conclusion**

The LA Airbnb market is a highly professional B2B (Business-to-Host) market, demanding high quality, and clearly divided between the volume market (Long Beach) and the premium market (Bel-Air/Malibu).

## **2. Recommendation : Pricing**

Competitive Pricing: For listings in volume areas, prices should stay within quartile 1–3 (\$107–\$228). Do not attempt to price above quartile 3 without verified Superhost status.

## **3. Recommendation : AI-Driven Marketing**

Qualitative Targeting: Use Granite classification results (e.g., 'Business Traveler') to format the top 10 amenities and listing descriptions to target guest segments willing to pay premium prices.

## **4. Recommendation : Host Quality**

Elite Benchmark: All new hosts should mirror the characteristics of 'Elite' hosts (100% Response Rate, Superhost) to enter the segment of 9,882 high-occupancy properties.



# *PERSONAL PROFILE*

*THANK  
YOU*

**Contact me! Klik here**



[nabila.hlwana@gmail.com](mailto:nabila.hlwana@gmail.com)



[nabila hulwana](#)



[nabilahlw](#)



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